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RESEARCH-ARTICLE

All you need is rotation: Construction of developable strips

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All you need is rotation: Construction of developable strips

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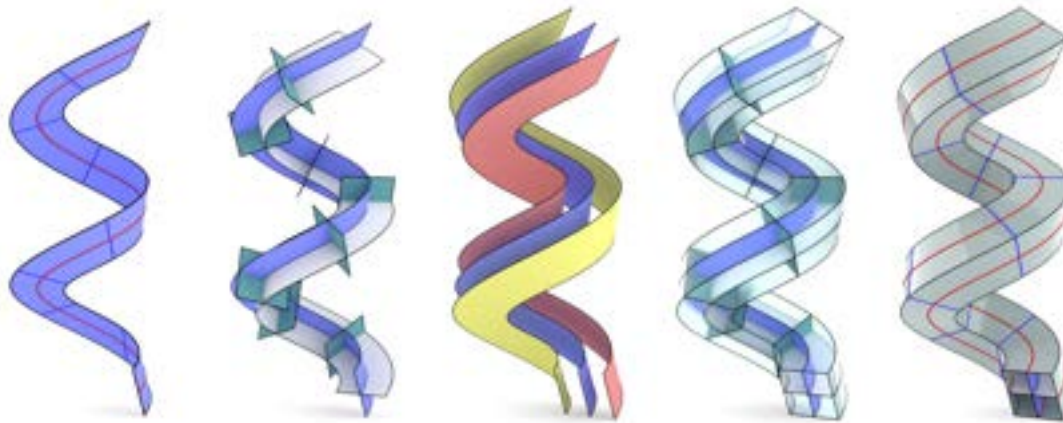


Fig. 1. Left to right: A developable strip is constructed along the rotation minimizing frame of a helix. An orthogonally intersecting developable strip is generated along the helix (depicted in silver) and the square plates (illustrated in green) are inserted along the rulings forming a triply orthogonal structure. The red and yellow developable offset surfaces neatly sandwich the central blue developable strip. Complementing this arrangement, a set of offset surfaces is positioned on either side of the silver strip, completing the triply orthogonal structure. The lines of curvature, reflecting both the directrix and rulings of the developable strips, trace across the encompassing four offset surfaces.

We present a novel approach to generate developable strips along a space curve. The key idea of the new method is to use the rotation angle between the Frenet frame of the input space curve, and its Darboux frame of the curve on the resulting developable strip as a free design parameter, thereby revolving the strip around the tangential axis of the input space curve. This angle is not restricted to be constant but it can be any differentiable function defined on the curve, thereby creating a large design space of developable strips that share a common directrix curve. The range of possibilities for choosing the rotation angle is diverse, encompassing constant angles, linearly varying angles, sinusoidal patterns, and even solutions derived from initial value problems involving ordinary differential equations. This enables the potential of the proposed method to be used for a wide range of practical applications, spanning fields such as architectural design, industrial design, and papercraft modeling. In our computational and physical examples, we demonstrate the flexibility of the method by constructing, among others, toroidal and helical windmill blades for papercraft models, curved foldings, triply orthogonal structures, and developable strips featuring a log-aesthetic directrix curve.

CCS Concepts: • **Computing methodologies** → **Parametric curve and surface models**.

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Additional Key Words and Phrases: developable strip, curved folding, toroidal and helical blades, triply orthogonal structure, log aesthetic curve

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1 Introduction

Developable surfaces are surfaces that can be unrolled onto a plane without stretching or tearing, making them highly desirable for manufacturing purposes. They form a special case of ruled surfaces, which are curved surfaces generated by the continuous motion of a straight line, known as a generator, along a space curve, referred to as a directrix. Developable surfaces are ruled surfaces with the additional property that the tangent plane remains constant along each generator.

1.1 Related work

It is known from classical differential geometry that developable surfaces can be constructed as the envelope of a family of planes along space curves [Willmore 2012], or of tangent planes along curves on a given input surface [do Carmo 1976]. The shape of the resulting developable surfaces is directly determined by the particular combination of curves and tangent planes selected for construction. Three types of tangent planes are commonly employed: osculating planes, normal planes, and rectifying planes. Each of these planes yields a distinct type of developable surface - osculating developable, polar

developable, and rectifying developable surfaces, respectively, see [Willmore 2012].

For the approximation of free-form surfaces with developable strips, a common approach is to use tangent planes along curves on the surface. As noted by [Scholz et al. 2023], this means that once these curves are determined, there is no further control over the ruling directions. This lack of control often leads to the emergence of edges of regression close to the directrix, which can result in exceedingly narrow developable strips.

One method based on this technique was proposed by [Raffaelli et al. 2018]. The authors choose curves on a given input surface and generate the unique developable surfaces that are tangent to the surface along these curves, a process they refer to as the ribbonization of the input surface. The success of this method heavily depends on the shape of the generated developable surfaces, which are fully defined by the choice of directrix curves along the input surface. In contrast, the techniques presented in our work provide greater flexibility in generating developable surfaces along a curve by relaxing the constraint that the developable surface must be tangent to the input surface, as illustrated in Section 7.5.

The history and applications of developable surfaces are outlined in [Lawrence 2011]. For a more recent and comprehensive overview, Yuan et al. provide a survey in [Yuan et al. 2023]. They classify the relevant literature into digital modeling, physical modeling, interaction, and practical applications.

[Seguin et al. 2021] investigate a family of developable surfaces that is generated from a space curve with curvature κ and a given function k satisfying $|k| \leq \kappa$. [Bulut and Callskan 2021] discuss the relationship between the spherical motion and developable surfaces with a directrix curve that is a line of curvature. In various optimization-based approaches [Kilian et al. 2008; Liu et al. 2006], the enhancement of developability and approximation quality is achieved through nonlinear optimization. Nevertheless, the presence of conflicting terms in the objective function in these methods can potentially compromise the overall developability of the obtained results. The problem of approximation with developable surfaces has been considered in [Sellán et al. 2020], where point clouds are approximated with developable height fields, resulting in a convex semidefinite optimization problem.

[Demaine et al. 2015] define curved folding as a hybrid process involving both folding and bending a sheet, resulting in a surface composed of curved creases and smooth developable surface patches. [Tachi 2011] provides equations that establish the connection between ruling directions and fold angles in a curved fold, building on the description found in [Fuchs and Tabachnikov 1999]. [Mitani and Ohashi 2023] propose a user interface that enables the interactive design of a developable surface with a curved fold along a user-specified space curve, by using an auxiliary curve called a "handle curve". [Jiang et al. 2019] investigate a pleated structure created by folding paper along a curved crease with a constant fold angle. Several recent works model developable surfaces within the framework of discrete differential geometry, where discrete notions of developability are determined that aim to mimic their smooth analogues. In particular discrete geodesic nets have been proposed as a flexible way to represent discrete developable surfaces as regular quadrilateral meshes [Rabinovich et al. 2018a,b; Wang

et al. 2019]. Discrete geodesic nets have also been applied to the problem of approximating arbitrary shapes with piecewise developable discrete surfaces [Ion et al. 2020]. Moreover, [Rabinovich et al. 2019] used them for modelling curved foldings. [Jiang et al. 2020] define discrete isometries via correspondences of checkerboard pattern on quadrilateral meshes. They then use these discrete isometries for geometric modeling, resulting in piecewise smooth developable spline surfaces. Besides quadrilateral meshes, also triangle meshes have been considered as a representation of discrete developable surfaces [Binninger et al. 2021; Stein et al. 2018]. The different definitions of discrete developability in these approaches result in optimization schemes that deform an arbitrary discrete input surface towards a discrete developable surface.

A key component of the approach presented in the present work is choosing a special moving frame along the input curve that facilitates finding a suitable developable surface. There are several other works that use adapted frames on curves and surfaces for different geometric applications. In [Bergou et al. 2008], the authors define a discrete Bishop frame along piecewise linear curves in order to model discrete elastic rods. [Coiffier and Corman 2023] consider the problem of finding a global parameterization of complicated surfaces via Cartan's method of moving frames. More precisely, they generalize the methods for discrete surfaces represented by triangle meshes, analyzed in [Lipman et al. 2007], to the case of singular frame fields. A method for parameterization of volumetric domains with hexahedral meshes based on Cartan's method of moving frames was presented by [Corman and Crane 2019].

1.2 Contribution

This paper presents a novel approach for generating developable strips along input space curves, providing greater flexibility in controlling their ruling directions. Our method involves defining the rotation angle φ between the Darboux frame of directrix curve on the resulting developable surface and the Frenet frame of the directrix curve as a space curve, revolving around the tangential axis of the space curve c as illustrated in Figure 2.

By utilizing the often overlooked potential of freely choosing the angle φ as a function defined on the directrix curve, we uncover its previously untapped capabilities. This strategic choice empowers us with refined control over the orientation of the rulings of the developable surfaces generated by our method. The range of possibilities for the rotation angle φ is diverse, spanning constant angles, linearly varying angles, sinusoidal patterns, and even solutions of ordinary differential equations. The proposed method finds versatile applications, including but not limited to architecture, papercraft windmill blade design, curved folding, triply orthogonal structures, and the creation of surfaces showcasing a log-aesthetic curve.

2 Frenet and Darboux frames

In this section, we review the differential geometry of the Frenet frame and the Darboux frame and their relation for curves on surfaces through the rotation angle φ .

2.1 Frenet frame

Let $\mathbf{c}(s)$ be an arc-length-parameterized space curve. We denote the unit tangent, principal normal, and the binormal vectors of $\mathbf{c}(s)$ by

$$\mathbf{t}(s) = \mathbf{c}'(s), \quad \mathbf{n}(s) = \frac{\mathbf{c}''(s)}{\|\mathbf{c}''(s)\|}, \quad \mathbf{b}(s) = \mathbf{t}(s) \times \mathbf{n}(s), \quad (1)$$

where the prime ' denotes differentiation with respect to the arc-length s and $\times$ denotes the cross product of two vectors. The curvature $\kappa(s)$ and torsion $\tau(s)$ of $\mathbf{c}(s)$ are given by:

$$\kappa(s) = |\mathbf{c}''(s)|, \quad \tau(s) = \frac{[\mathbf{c}'(s), \mathbf{c}''(s), \mathbf{c}'''(s)]}{\mathbf{c}''(s) \cdot \mathbf{c}''(s)}, \quad (2)$$

where $[\cdot, \cdot, \cdot]$ denotes the scalar triple product. For an arbitrarily parameterized curve $\mathbf{c}(t)$, the unit tangent, principal normal, and the binormal vectors of $\mathbf{c}(t)$ are given by

$$\mathbf{t}(t) = \frac{\dot{\mathbf{c}}(t)}{\|\dot{\mathbf{c}}(t)\|}, \quad \mathbf{b}(t) = \frac{\dot{\mathbf{c}}(t) \times \ddot{\mathbf{c}}(t)}{\|\dot{\mathbf{c}}(t) \times \ddot{\mathbf{c}}(t)\|}, \quad \mathbf{n}(t) = \mathbf{b}(t) \times \mathbf{t}(t), \quad (3)$$

where dot $\dot{\cdot}$ denotes the differentiation with respect to the parameter t . The curvature $\kappa(t)$ and torsion $\tau(t)$ are given by

$$\kappa(t) = \frac{\|\dot{\mathbf{c}}(t) \times \ddot{\mathbf{c}}(t)\|}{\|\dot{\mathbf{c}}(t)\|^3}, \quad \tau(t) = \frac{[\dot{\mathbf{c}}(t), \ddot{\mathbf{c}}(t), \ddot{\mathbf{c}}(t)]}{(\dot{\mathbf{c}}(t) \times \ddot{\mathbf{c}}(t)) \cdot (\dot{\mathbf{c}}(t) \times \ddot{\mathbf{c}}(t))}. \quad (4)$$

The following differential equations in matrix form, known as Frenet-Serret formulae [Struik 2012],

$$\begin{bmatrix} \dot{\mathbf{t}} \\ \dot{\mathbf{n}} \\ \dot{\mathbf{b}} \end{bmatrix} = \begin{bmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{bmatrix}, \quad (5)$$

describe the motion of the moving orthonormal triplet $(\mathbf{t}, \mathbf{n}, \mathbf{b})$ along the curve $\mathbf{c}(s)$. For an arbitrarily parameterized curve the Frenet-Serret formulae are given by

$$\begin{bmatrix} \dot{\mathbf{t}} \\ \dot{\mathbf{n}} \\ \dot{\mathbf{b}} \end{bmatrix} = \Lambda(t) \begin{bmatrix} 0 & \kappa(t) & 0 \\ -\kappa(t) & 0 & \tau(t) \\ 0 & -\tau(t) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{bmatrix}, \quad (6)$$

where $\Lambda(t) = \frac{ds}{dt} = \|\dot{\mathbf{c}}(t)\|$ is the parametric speed.

2.2 Darboux frame

Curves on surfaces possess an additional natural orthogonal frame. Let $\mathbf{R}(u, v)$ be a smooth parametric surface in $\mathbb{R}^3$ and let $\mathbf{c}(s) = \mathbf{R}(u(s), v(s))$ be an arc-length-parameterized curve on the surface. The Darboux frame $(\mathbf{t}, \mathbf{B}, \mathbf{N})$ [do Carmo 1976; Gray 1999] is an orthogonal frame for the curve $\mathbf{c}$, similar to the Frenet frame (see Fig. 2). It is often employed in the motion planning of autonomous vehicles [Shiller and Gwo 1991; Usami et al. 2020] and a camera mounted on a moving platform [Duric and Rosenfeld 2003]. Note that in the computer vision field, the local coordinate system comprising the surface normal at a point and the two principal directions is also referred to as the Darboux frame [Hameiri and Shimshoni 2003]. In our research the Darboux frame is evaluated as

$$\mathbf{t} = \frac{\mathbf{R}_u u' + \mathbf{R}_v v'}{\|\mathbf{R}_u u' + \mathbf{R}_v v'\|}, \quad \mathbf{N} = \frac{\mathbf{R}_u \times \mathbf{R}_v}{\|\mathbf{R}_u \times \mathbf{R}_v\|}, \quad \mathbf{B} = \mathbf{N} \times \mathbf{t},$$

where $\mathbf{t}$ is the unit tangent vector of $\mathbf{c}$ and $\mathbf{N}$ is the unit normal vector of the parametric surface $\mathbf{R}$. Given that $\mathbf{t}, \mathbf{B}$, and $\mathbf{N}$ are unit vectors, it follows that $\mathbf{t} \cdot \mathbf{t}' = 0$, $\mathbf{B} \cdot \mathbf{B}' = 0$, and $\mathbf{N} \cdot \mathbf{N}' = 0$. Consequently,

we can infer that $\mathbf{t}'$, $\mathbf{B}'$, and $\mathbf{N}'$ are situated within the $\mathbf{BN}$ -plane, $\mathbf{tN}$ -plane, and $\mathbf{tB}$ -plane, respectively. This leads to

$$\mathbf{t}' = c_1 \mathbf{B} + c_2 \mathbf{N}, \quad \mathbf{B}' = c_3 \mathbf{t} + c_4 \mathbf{N}, \quad \mathbf{N}' = c_5 \mathbf{t} + c_6 \mathbf{B}. \quad (7)$$

The coefficients of $\mathbf{t}'$ are called geodesic curvature and normal curvature, respectively, and we denote them by $c_1 = \kappa_g$ and $c_2 = \kappa_n$. The coefficient of $\mathbf{B}'$ with respect to $\mathbf{N}$ is called geodesic torsion, see [Kreyszig 1968], and is denoted by $c_4 = \mathbf{B}' \cdot \mathbf{N} = -\mathbf{B} \cdot \mathbf{N}' = [\mathbf{N}, \mathbf{N}', \mathbf{t}] = \tau_g$. For the remaining coefficients we obtain

$$c_3 = \mathbf{B}' \cdot \mathbf{t} = -\kappa_g, \quad c_5 = \mathbf{N}' \cdot \mathbf{t} = -\kappa_n, \quad c_6 = \mathbf{B} \cdot \mathbf{N}' = -\tau_g.$$

We can thus rewrite (7) in matrix form, yielding the motion of the Darboux frame:

$$\begin{bmatrix} \dot{\mathbf{t}} \\ \dot{\mathbf{B}} \\ \dot{\mathbf{N}} \end{bmatrix} = \begin{bmatrix} 0 & \kappa_g(s) & \kappa_n(s) \\ -\kappa_g(s) & 0 & \tau_g(s) \\ -\kappa_n(s) & -\tau_g(s) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ \mathbf{B} \\ \mathbf{N} \end{bmatrix}. \quad (8)$$

For arbitrary speed curve Eqn. (8) formulae are given by

$$\begin{bmatrix} \dot{\mathbf{t}} \\ \dot{\mathbf{B}} \\ \dot{\mathbf{N}} \end{bmatrix} = \Lambda(t) \begin{bmatrix} 0 & \kappa_g(t) & \kappa_n(t) \\ -\kappa_g(t) & 0 & \tau_g(t) \\ -\kappa_n(t) & -\tau_g(t) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ \mathbf{B} \\ \mathbf{N} \end{bmatrix}. \quad (9)$$

2.3 Relation between Frenet and Darboux frames

Since the tangent vectors of the Frenet frame (5) and the Darboux frame (8) are the same, as shown in Figure 2, there is an angle φ such that a rotation of $\mathbf{n}$ and $\mathbf{b}$ in the normal plane produces $\mathbf{B}$ and $\mathbf{N}$:

$$\begin{bmatrix} \mathbf{t} \\ \mathbf{B} \\ \mathbf{N} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{bmatrix}. \quad (10)$$

If we plug (10) into (8), the second and third equations of (8) become

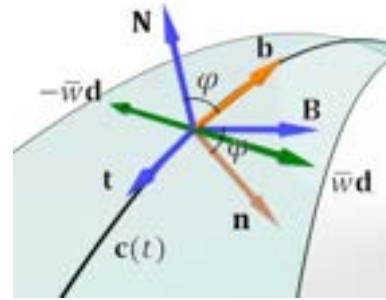


Fig. 2. The relation between the Frenet frame and the Darboux frame. The ruling direction $\mathbf{d}$ is determined by (17).

$$\mathbf{B}' = -\kappa_g \mathbf{t} - \tau_g \sin \varphi \mathbf{n} + \tau_g \cos \varphi \mathbf{b}, \quad (11)$$

$$\mathbf{N}' = -\kappa_n \mathbf{t} - \tau_g \cos \varphi \mathbf{n} - \tau_g \sin \varphi \mathbf{b}. \quad (12)$$

We can also differentiate the second and third equations of (10) yielding

$$\mathbf{B}' = -\kappa \cos \varphi \mathbf{t} - \sin \varphi \left(\tau + \frac{d\varphi}{ds} \right) \mathbf{n} + \cos \varphi \left(\tau + \frac{d\varphi}{ds} \right) \mathbf{b}, \quad (13)$$

$$\mathbf{N}' = \kappa \sin \varphi \mathbf{t} - \cos \varphi \left(\tau + \frac{d\varphi}{ds} \right) \mathbf{n} - \sin \varphi \left(\tau + \frac{d\varphi}{ds} \right) \mathbf{b}. \quad (14)$$



Fig. 3. Generation of a developable surface. Left: Two endpoints of the rulings are evaluated for fitting. Right: Fitted B-spline developable surface where white circles indicate the control points.

By comparing (11) and (13), (12) and (14) and we can establish the following relationships:

$$\kappa_g = \kappa \cos \varphi, \quad \tau_g = \tau + \frac{d\varphi}{ds}, \quad \kappa_n = -\kappa \sin \varphi. \quad (15)$$

These equations indicate that κ_n , κ_g , and τ_g can be derived from the curvature and torsion of the space curve, in conjunction with the rotation angle $\varphi(s)$ between the Darboux frame and the Frenet frame.

3 Ruling direction, representation, and unfolding

In this section, we study the derivation of the ruling direction vector of developable surfaces in two scenarios: one where the input curve is provided and another where the curvature and torsion are specified instead of the curve itself. Subsequently, we discuss the control over the width of developable strips, which enhances their design, and introduce an effective approach for representing developable strips through B-spline fitting. Lastly, we outline a method for unfolding these developable strips onto a plane.

3.1 Ruling direction from input curves

3.1.1 Conventional method. Given a curve $\mathbf{c}$ on a surface, the ruling direction of a developable surface can be obtained by taking the envelope of the family of tangent planes along the curves [do Carmo 1976; Scholz and Maekawa 2021; Scholz et al. 2023; Takezawa et al. 2021]. According to this definition, the ruling direction becomes parallel to $\mathbf{N} \times \mathbf{N}'$, and the unit ruling direction can be obtained using the third equation of (8):

$$\mathbf{d} = \frac{\tau_g \mathbf{t} - \kappa_n \mathbf{B}}{\sqrt{\tau_g^2 + \kappa_n^2}}. \quad (16)$$

Once the curves on surfaces are fixed, the ruling direction is automatically determined.

However, the lack of control over the ruling direction often leads to the appearance of an edge of regression, also known as a cuspidal edge, near $\mathbf{c}$, potentially leading to a narrow developable strip. In Section 4.3, we illustrate how the emergence of the edge of regression can be controlled by adjusting the rotation angle φ . Furthermore, such examples are given in Sections 5.3 and 7.4.

3.1.2 Proposed method. In this paper, we present a novel solution to address this issue by generating developable strips along space curves, including curves on surfaces. Let $\mathbf{c}$ be an arbitrary smooth space curve. In order to design a developable strip along this curve, we freely choose a rotation angle φ as a function defined on $\mathbf{c}$. We

then find a developable strip that contains $\mathbf{c}$ such that φ is the rotation angle between the Frenet frame of $\mathbf{c}$ and the Darboux frame of $\mathbf{c}$ regarded as curve on the developable strip.

To find this surface, we calculate $\mathbf{t}$, $\mathbf{n}$, and $\mathbf{b}$ using (1) or (3), followed by the derivation of $\mathbf{B}$ using the second equation of (10). Employing (15), we are able to compute the ruling direction (16) as follows:

$$\mathbf{d}(\varphi) = \frac{(\tau + \frac{d\varphi}{ds})\mathbf{t} + \kappa \sin(\varphi) \cos(\varphi)\mathbf{n} + \kappa \sin^2(\varphi)\mathbf{b}}{\sqrt{(\tau + \frac{d\varphi}{ds})^2 + \kappa^2 \sin^2(\varphi)}}. \quad (17)$$

Note that the direction of the generators is reversed before and after an inflection point. However, this does not hinder us from calculating the two endpoints of the generators. In cases where the generator at a sample point $\mathbf{c}_i$ coincides exactly with an inflection point, we can simply exclude that specific point from our computation. Unlike the conventional method for generating developable surfaces along curves on surfaces, where the ruling directions of the resulting developable surfaces are inherently fixed, we thus introduced an additional degree of adaptability. This is achieved by incorporating the user-provided rotation angle φ and, consequently, $\frac{d\varphi}{ds}$ into the determination of the ruling directions (17). In Section 4.3, we explore how adjusting φ impacts the edge of regression curve.

Notably, for generating the tangential developable surface, φ is set to 0, and for generating the rectifying developable surface, φ is set to $\frac{\pi}{2}$. The new method generalizes the conventional method described in Section 3.1.1, which can be obtained by setting φ to $\cos^{-1}(\mathbf{b} \cdot \mathbf{N})$.

Another interesting choice is to enforce the Darboux frame to be a rotation minimizing frame [Wang et al. 2008], which is characterized by $\tau + \frac{d\varphi}{ds} = 0$. This choice ensures that $\mathbf{c}$ is a line of curvature and that the rulings are orthogonal to $\mathbf{t}$. In Section 5.1, we will consider the more general case of constant angles between the tangent vector of $\mathbf{c}$ and the rulings of the developable strip.

A similar equation to (17) is derived in [Jiang et al. 2019]. It's worth mentioning that in that study, they assume a constant value for φ (i.e. $d\varphi/ds = 0$), which leads to a constrained set of developable surfaces.

The exhaustive discussion of explicit choices of φ can be found in Section 4, while the scenario of determining the curve from provided curvature, torsion, and the rotation angle specified by the user is studied in the next section.

3.2 Finding the ruling direction by integrating the Darboux frame

In some situations, we are provided with the curvature and the torsion instead of the curve itself. In such cases, we first determine φ and evaluate normal curvatures κ_n , geodesic curvatures κ_g , and geodesic torsions τ_g using (15). Then, we set up a system of coupled nonlinear ordinary equations from (9) with $\dot{\mathbf{c}}(t) = \Lambda(t)\mathbf{t}(t)$ and appropriate initial values.

If we denote the initial values for the Frenet frame of the space curve as $\mathbf{c}(0)$ and $\mathbf{t}(0)$, $\mathbf{n}(0)$, and $\mathbf{b}(0)$, initial values for the Darboux frame can be obtained by rotating the Frenet frame by φ in the

normal plane using (10):

$$\mathbf{B}(0) = \cos \varphi \mathbf{n}(0) + \sin \varphi \mathbf{b}(0), \quad \mathbf{N}(0) = -\sin \varphi \mathbf{n}(0) + \cos \varphi \mathbf{b}(0).$$

Based on these initial values, we can numerically solve the initial value problem to obtain $\mathbf{c}(t_i)$, $\mathbf{t}(t_i)$, $\mathbf{B}(t_i)$, $\mathbf{N}(t_i)$ at discrete points, where $i = 0, \dots, n$. We employed the Runge-Kutta method to solve this system of equations. Finally, we obtain the ruling direction using (16).

An example where this procedure is useful is presented in Section 6, where we generate a developable strip that highlights a log-aesthetic curve [Miura 2006; Yoshida et al. 2009].

3.3 Controlling of width of developable strips

After determining the ruling direction at equally spaced points along the input curve $\mathbf{c}_i$ ($i = 0, \dots, n$), we calculate the two endpoints of the generator at each discrete point, as shown in Figure 3 (left), using the following expressions:

$$\mathbf{p}_i = \mathbf{c}_i + \bar{w}_i \mathbf{d}_i, \quad \mathbf{m}_i = \mathbf{c}_i - \bar{w}_i \mathbf{d}_i. \quad (18)$$

Given that $\mathbf{d}$ is generally not orthogonal to the input space curve, we calculate $\bar{w}_i$ as $w/|\mathbf{d}_i \cdot \mathbf{B}_i|$, where $w > 0$ represents half the width of the desired developable strip, ensuring that the strip maintains a width of w . In some cases, using different widths for the two sides can be beneficial, as shown in Section in Figure 9. More generally, we can assign the widths of the sides of the developable surface using appropriate functions $\bar{w}_p(t)$ and $\bar{w}_m(t)$. This feature highlights one of the strengths of our method, which relies on the ruling-based construction of developable strips. In this way, we do not need to trim the resulting developable strips, which simplifies downstream applications. Further examples illustrating the control of strip width can be found in Sections 7.1 and 7.4.

3.4 Representation of developable strips by B-splines

While representing developable surfaces with meshes is an option, we opted for the B-spline representation for developable strips because it aligns the ruling direction with the isoparametric lines, facilitating visualization as illustrated in Figure 4 (top right). We assign parameter values $0 \leq t_i \leq 1$ ($i = 0, \dots, n$) to each $\mathbf{c}_i$. The choice between the chord length and centripetal methods depends on the nature of the input data. When the data exhibits sharp turns, the centripetal method yields better results than the chord length method [Bingol and Krishnamurthy 2019; Piegl and Tiller 1997]. B-spline curve fitting is conducted on the two datasets, designated as $\mathbf{p}_i$ and $\mathbf{m}_i$, utilizing the parameterization of $\mathbf{c}_i$. The number of control points is set to 20 % of the number of points (Npt) that were sampled from the curve, with the internal knots determined through the methodology outlined in section 9.4.1 of [Piegl and Tiller 1997]. The resulting fitted curves are denoted as $\mathbf{p}(t)$ and $\mathbf{m}(t)$ for the plus and minus sides, respectively. It's worth noting that, unlike many optimization-based methods, our approach ensures precision in computing the generators, with any errors stemming exclusively from this approximation. Finally, we employ linear interpolation between $\mathbf{m}(t)$ and $\mathbf{p}(t)$ to generate the developable strip.

We evaluate the developability of the resulting models using a measure introduced in the reference [Pottmann et al. 2008]. This measure assesses the coplanarity of four points: $\mathbf{p}$, $\mathbf{m}$, $\mathbf{p} + \bar{\mathbf{p}}$, and

$\mathbf{m} + \bar{\mathbf{m}}$. This evaluation can be further translated into computing the distance between two diagonal lines connecting $\mathbf{p}$ and $\mathbf{m} + \frac{|\mathbf{p}-\mathbf{m}|}{|\bar{\mathbf{m}}|} \bar{\mathbf{m}}$, and $\mathbf{m}$ and $\mathbf{p} + \frac{|\mathbf{p}-\mathbf{m}|}{|\bar{\mathbf{p}}|} \bar{\mathbf{p}}$, as illustrated in the top left image in Figure 4. The top right image shows two B-spline developable strips, each with 120 control points, constructed around the helix with generator lengths of 10 (in blue) and 100 (in gray). The bottom graph in Figure 4 illustrates how the maximum and average distances between diagonal lines, expressed as a percentage of the average diagonal length, denoted by δ , increase as the generator length extends from 10 to 100 for the surfaces shown in the top right image. This occurs because, with the increasing length of the ruling, the distributions of $\mathbf{p}_i$ and $\mathbf{m}_i$ cease to accurately mirror the parametrization of $\mathbf{c}_i$, leading to a corresponding increase in fitting errors. The graph shows that the maximum distance between the diagonals (δ), expressed as a percentage of the average diagonal length, is on the order of 10^{-4} .

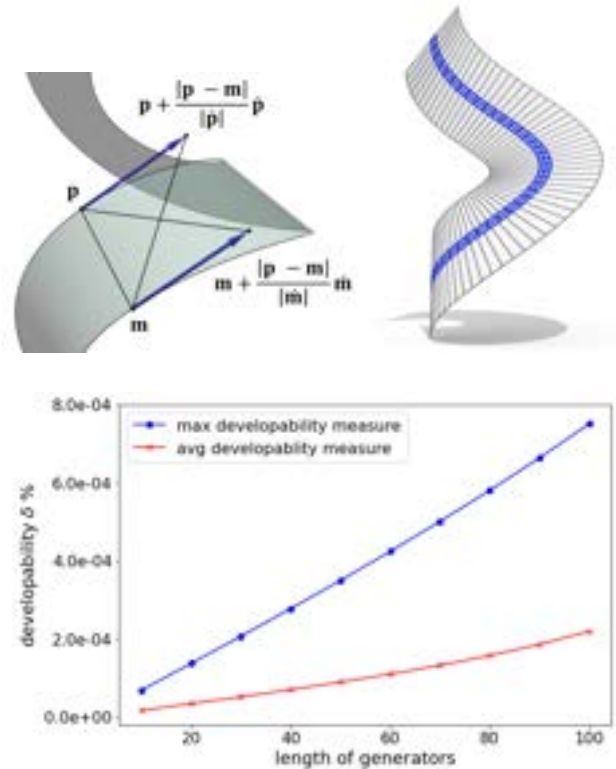


Fig. 4. Developability measure δ : Top left, the definition of developability measure, Top right, two B-spline developable strips constructed around the helix with generator lengths of 10 (in blue) and 100 (in gray), Bottom, the maximum and average developability measures (δ) of the strips in the top right image increase as the generator length extends from 10 to 100. The helix is sampled with Npt = 300.

3.5 Unfolding of developable strips

Given that developable strips have the property of being unfoldable onto a plane without any need for stretching or tearing, we present a simple and straightforward technique to seamlessly flatten these

strips onto a plane. This enables us to effectively employ this method for various fabrication purposes. A more sophisticated method which involves the solution of ordinary differential equation is described in Section 7.3.

Elaborating on the construction process outlined in Section 3.4, we begin with a pair of endpoints for each ruling, denoted as $\mathbf{m}_i$ and $\mathbf{p}_i$, which are positioned exactly on the developable strip with a high level of precision. This concept is visually represented in Figure 5, where we examine two rulings, $\mathbf{m}_i - \mathbf{p}_i$ and $\mathbf{m}_{i+1} - \mathbf{p}_{i+1}$, represented as AD and BC, respectively. We begin by splitting the quadrilateral ABCD into two triangles, ABD and ACD, while ensuring that the original edge lengths (AD, AB, and CD) and angles (ξ_A and ξ_D) are preserved, as depicted in Figure 5 (Top left and right).

The resulting flattened two triangles are denoted as $\bar{A}\bar{B}\bar{D}$ and $\bar{A}\bar{C}\bar{D}$, where $\bar{A}$ is defined as the origin and the edge $\bar{A}\bar{D}$ aligns with the horizontal axis. At $i=0$, we set the rotation angle $\xi=0$ and the translation vector as $\mathbf{T}=(0,0)$. We then apply the transformation $(\xi, \mathbf{T})$ to $\bar{B}$ and $\bar{C}$, resulting in $\bar{B}_g$ and $\bar{C}_g$, respectively, and store these as flattened data. Next, we calculate the incremental rotation angle $\Delta\xi$ of the edge $\bar{B}\bar{C}$ relative to the horizontal axis. We update ξ by $\xi=\xi + \Delta\xi$ and set $\mathbf{T}=\bar{\mathbf{B}}_g$, incrementing i by 1. This iterative procedure is applied to all quadrilaterals from the bottom and, as illustrated in Figure 5 (bottom left). This approach preserves the angles ξ_A^i at each iteration.

The source of error arises when treating AB and CD as line segments, which becomes negligible with an increasing number of generators. Figure 5 (bottom right) displays the unfolded toroidal blade developable strip from Figure 17, exhibiting symmetry with respect to the centerline when 56 line segments are used, confirming the negligible nature of the error. This flattening algorithm is employed in Section 7 for unfolding developable strips for fabrication.

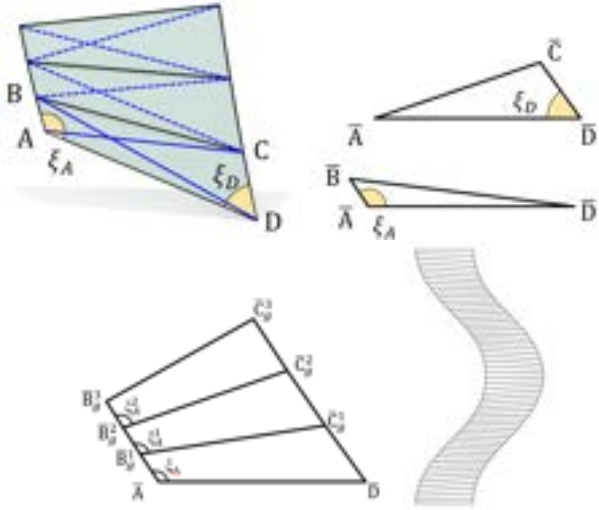


Fig. 5. Flattening of developable strip. Top left: Developable strip in 3D. Top right: flattened two triangles. Bottom left: flattened developable strip. Bottom right, flattened toroidal blade developable strip of Figure 17.

4 Arranging explicit rotation angles

In this section, we examine different examples for explicit rotation angles for planar and space curves in the construction of the developable strips. Afterwards, in Section 5, we explore rotation angles obtained by solving ordinary differential equations.

4.1 Planar curves

4.1.1 Constant rotation angle. We initiate our investigation on the proposed method using an ellipse, defined as follows:

$$\mathbf{c}(t) = (a \cos(t), b \sin(t), 0).$$

Its parametric speed and the curvature can be expressed as

$$\Lambda(t) = \sqrt{a^2 \sin^2(t) + b^2 \cos^2(t)}, \quad \kappa(t) = \frac{ab}{\sqrt{a^2 \sin^2(t) + b^2 \cos^2(t)}}.$$

If we set the rotation angle to $\varphi = \frac{\pi}{2}$, the values of κ_g and κ_n become 0 and $-\kappa$, respectively. This results in the formation of a cylindrical rectifying developable surface. Figure 6 visually represents the variations in the shape of the developables as the constant rotation angle φ transitions through $\frac{\pi}{2}$, $\frac{\pi}{3}$, $\frac{\pi}{6}$, and $\frac{\pi}{9}$ from top left to bottom right. In this example, we used $a=3$, $b=1.5$ for $0 \leq t \leq 2\pi$, and $w=0.5$. As we decrease φ , we can observe a flattening of the developables, which corresponds to an increase in geodesic curvature.

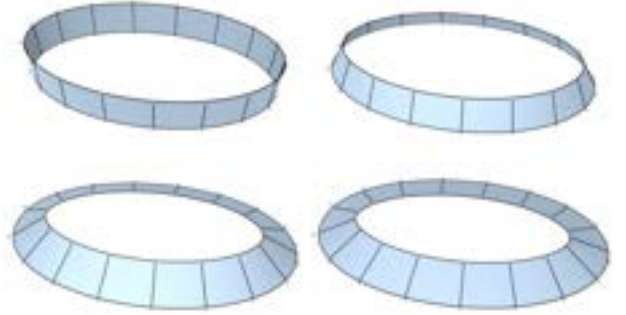


Fig. 6. Variations in the shape of the developables as the fixed rotation angle φ transitions through $\frac{\pi}{2}$, $\frac{\pi}{3}$, $\frac{\pi}{6}$, and $\frac{\pi}{9}$ from the top left to the bottom right. The circles represent the control points of the B-spline developable strip. The ellipse is sampled with $N_{pt} = 100$.

4.1.2 Parameter-dependent rotation angle. We enhance the constant rotation angle φ by incorporating a function of the curve parameter t . By setting $\varphi(t) = 0.3 \sin(2t) + \frac{\pi}{2}$, intriguing changes occur in the geodesic torsion τ_g . Specifically, while the geodesic torsion for constant rotation angles vanishes, it is evaluated as $\frac{d\varphi}{ds} = \frac{1}{\Lambda(t)} \frac{d\varphi}{dt} = \frac{0.6 \cos(2t)}{\Lambda(t)}$ for this angle function.

To provide a visual understanding, Figure 7 top images distinctly depict contrasting between the top views of the resulting developable strips using $\varphi = \frac{\pi}{2}$ and the parameter-dependent rotation angle $\varphi(t) = 0.3 \sin(2t) + \frac{\pi}{2}$. These views provide a clear illustration of the impact caused by incorporating the parameter-dependent function $0.3 \sin(2t)$ into the constant rotation angle. Moreover, the two bottom images in the figure present the front and perspective

views, respectively. This comprehensive visual representation allows for a more complete understanding of the effects resulting from the addition of the parameter-dependent function to the fixed rotation angle $\frac{\pi}{2}$.

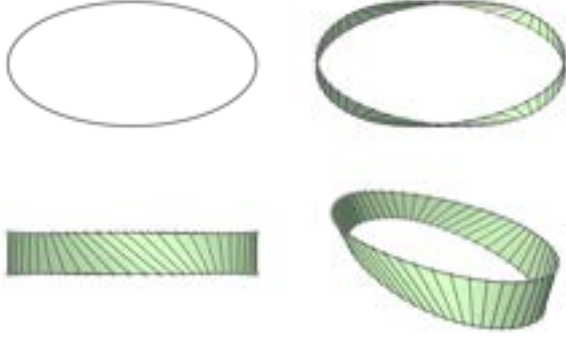


Fig. 7. The two top left and right images distinctly depict the contrasting top views between $\varphi = \frac{\pi}{2}$ and $\varphi(t) = 0.3 \sin(2t) + \frac{\pi}{2}$, respectively. The bottom left and right images present the front and perspective views, respectively. The ellipse is sampled with $N_{pt} = 100$.

4.2 Space curves

4.2.1 *Helix*. A circular helix in parametric representation is given by

$$\mathbf{c}(t) = (a \cos(t), a \sin(t), bt),$$

with constant parametric speed, curvature, and torsion:

$$\Lambda = \sqrt{a^2 + b^2}, \quad \kappa = \frac{a}{a^2 + b^2}, \quad \tau = \frac{b}{a^2 + b^2}.$$

We can calculate the unit tangent, principal normal, and binormal vectors of $\mathbf{c}(t)$ using the equation (4). Once we have determined $\varphi(t)$, we are then prepared to compute the unit ruling direction vector $\mathbf{d}$ as described in equation (17).

In Figure 8, the top row illustrates, from left to right, the top view, front view, perspective view of the developable strip associated with a helix ($a = 3$, $b = 1$, $w = 1.5$), having a fixed rotation angle of $\frac{\pi}{2}$ and parameter range $0 \leq t \leq 4\pi$. This strip represents a rectifying developable surface. In contrast, the bottom row displays the corresponding views for a fixed rotation angle of $\frac{\pi}{6}$.

In the following example, we showcase an architectural design that utilizes a helix with $a = 20$ and $b = 20$, spanning the parameter range $0 \leq t \leq 2\pi$. While the common practice in architectural design is to apply a tangential developable surface ($\varphi = 0$) [Martin-Pastor and López-Martínez 2020], our design deviates by employing $\varphi(t) = -0.06t + \frac{\pi}{6}$. We then determine the lengths of the plus and minus rulings as follows:

$$\mathbf{p}_i = \mathbf{c}_i, \quad \mathbf{m}_i = \mathbf{c}_i - \frac{z(t)}{d_z} \mathbf{d}_i, \quad (19)$$

where d_z is the z-component of the ruling direction vector (17). In Figure 9, the sequence from top left to bottom right illustrates the perspective view, top view, flattened data, and the papercraft model

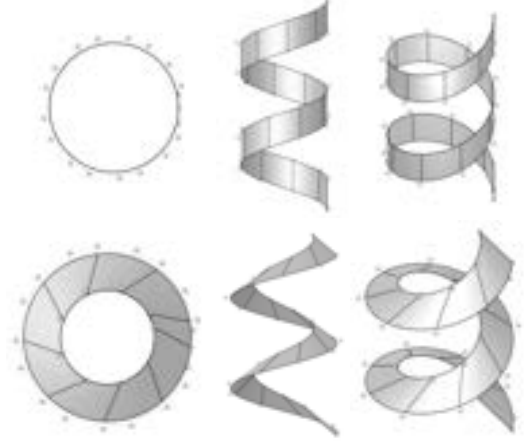


Fig. 8. The top row illustrates, from left to right, the top view, front view, and the perspective view of the developable strip helix with fixed rotation angle $\frac{\pi}{2}$. In contrast, the bottom row displays the corresponding views for a fixed rotation angle of $\frac{\pi}{6}$. The helix is sampled with $N_{pt} = 100$.



Fig. 9. Architectural model. From the top left to the bottom right, perspective view, top view, flattened data, and papercraft model of the architectural design. The helix is sampled with $N_{pt} = 100$.

of this architectural design. The impact of varying the function $\varphi(t)$ is further visualized in the supplemental video, where we show the change of the developable surface along the helix curve generated by $\varphi(t) = -0.06t + q$ for $q \in [(\frac{1}{6} - \frac{1}{2})\pi, \frac{\pi}{6}]$ in an animation.

4.2.2 *Möbius strip*. A Möbius strip is a non-orientable surface that can be created by taking a strip of paper, giving it a half-twist, and

then connecting its ends to form a loop. [Schwarz 1990] generates an Möbius strip as a rectifying developable ($\varphi = \frac{\pi}{2}$) of the algebraic curve given in parametric form by

$$\mathbf{c}(t) = (\sin(t), (1 - \cos(t))^3, \sin(t)(1 - \cos(t))) . \quad (20)$$

The curvature and torsion can be computed using (3). The top left

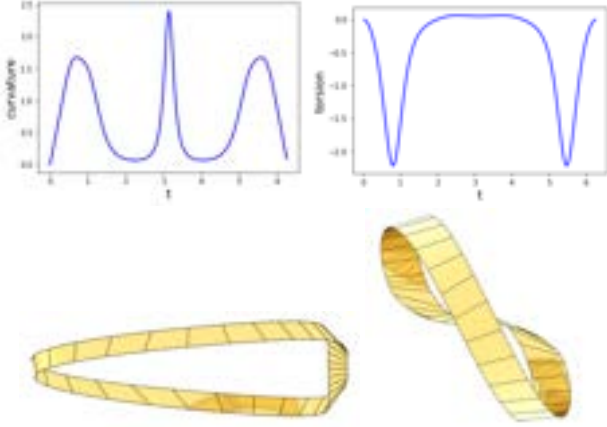


Fig. 10. Möbius strip. Top left to bottom right, plot of curvature, plot of torsion, top view, and perspective view. The directrix curve (20) is sampled with $N_{pt} = 200$.

and the right graphs in Figure 10 show the plots of curvature and torsion of the space curve (20), respectively. The bottom left and right images depict the top and perspective views, respectively, of the resulting Möbius strip.

4.3 Control of edge of regression

In contrast to conventional methods for constructing developable surfaces, which lack the ability to control the emergence of the edge of regression, our method allows us to control this emergence by adjusting the rotation angle φ . The edge of regression $\mathbf{e}$ of the developable surface is defined by [Scholz et al. 2023]:

$$\mathbf{e}(t) = \mathbf{c}(t) - \Lambda \kappa_n \frac{\tau_g \mathbf{t} - \kappa_n \mathbf{B}}{\Lambda \kappa_g (\kappa_n^2 + \tau_g^2) + \dot{\tau}_g \kappa_n - \dot{\kappa}_n \tau_g} . \quad (21)$$

Figure 11 illustrates how the edge of regression curve changes as the rotation angle φ along the helix varies, with $a = 20$ and $b = 10$. The rotation angles are defined as $\varphi = -0.06t + q$, where q takes values of $\frac{\pi}{4}$ (45°), $\frac{5\pi}{18}$ (50°), $\frac{11\pi}{36}$ (55°), and $\frac{\pi}{3}$ (60°). The left image shows the edge of regression curve of the helix for $q = \frac{\pi}{4}$. As q increases, the length of $\mathbf{e}(t)$ also increases, as depicted in the graph (center), along with the top views of the four developable strips (right). We provide two examples of avoiding the emergence of the edge of regression in Sections 5.3 and 7.4.

5 Rotation angles that result from solving an ODE

In this section, we analyze different schemes where, instead of defining it explicitly, the rotation angle φ is given as the solution of an initial value problem.

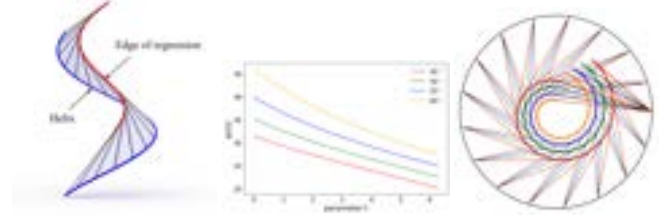


Fig. 11. Edge of regression: From left to right, $q = \frac{\pi}{4}$, graph of edge of regression, and top views of the four developable strips. The helix is sampled with $N_{pt} = 100$.

5.1 Developable strips with constant angle between tangent and ruling direction

We define the constant angle between the tangent vector $\mathbf{t}$ of the space curve and the ruling direction vector $\mathbf{d}$ as θ . When we take the dot product of both sides of equation (17) with the unit tangent vector $\mathbf{t}$, we obtain:

$$\mathbf{t} \cdot \mathbf{d} = \frac{\tau + \frac{d\varphi}{ds}}{\sqrt{(\tau + \frac{d\varphi}{ds})^2 + \kappa^2 \sin^2(\varphi)}} = \cos(\theta) = \text{constant} . \quad (22)$$

This relationship can be reformulated into the shape of a first-order ordinary differential equation:

$$\frac{d\varphi}{ds} = \pm \frac{\kappa \sin(\varphi)}{\tan(\theta)} - \tau . \quad (23)$$

The $\pm$ comes from the fact that the sign of θ is not relevant to the value of $\cos \theta$. Multiplying equation (22) with the denominator of the center equation, it is obvious that $\tau + \frac{d\varphi}{ds} = \tau_g > 0$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Thus, equation (23) indicates that $\pm \frac{\kappa \sin(\varphi)}{\tan(\theta)}$ is positive. For an arbitrary speed parameter, it becomes:

$$\frac{d\varphi}{dt} = \Lambda \left(\pm \frac{\kappa \sin(\varphi)}{\tan(\theta)} - \tau \right) .$$

This equation can be integrated as an initial value problem using standard numerical techniques like the Runge-Kutta method. We used the initial condition $\varphi(0) = 0$. It's worth noting that equation (22) remains valid even if we add a constant angle $\hat{\varphi}$ to $\varphi(t)$. In other words, if we substitute

$$\frac{d\varphi}{dt} = \Lambda \left(\pm \frac{\kappa \sin(\varphi + \hat{\varphi})}{\tan(\theta)} - \tau \right)$$

into (22), it will yield the right-hand side. In the special case where the tangent of the space curve and ruling directions are orthogonal ($\theta = \frac{\pi}{2}$), denoted as $\mathbf{t} \cdot \mathbf{d} = 0$, this leads to $\tau + \frac{d\varphi}{ds} = 0$ as a consequence of equation (22). Essentially, this condition implies that the geodesic torsion τ_g becomes zero. Hence, the computation of the rotation angle simplifies to a straightforward integration problem, derived from the second equation of (15), given by:

$$\varphi(t) = - \int_{t_0}^t \Lambda \tau(t) dt + \varphi_0 . \quad (24)$$

The moving frame with $\tau_g = 0$ is known as a rotation minimizing frame (RMF) [Wang et al. 2008]. By using the rotation angle (24) along the space curve $\mathbf{c}(t)$, the space curve itself becomes a line

of curvature for the resulting developable surface. This is directly linked to the intriguing property that the geodesic torsion of a line of curvature becomes zero, as stated in [Kreyszig 1968; Pan et al. 2015]. Furthermore, all the rulings become the lines of curvature with zero principal curvature [Maekawa and Chalfant 1998]. We exemplify our method by presenting helical developable surfaces in Figure 12, progressing from top left to bottom right, with θ values of $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, and $\frac{\pi}{2}$.

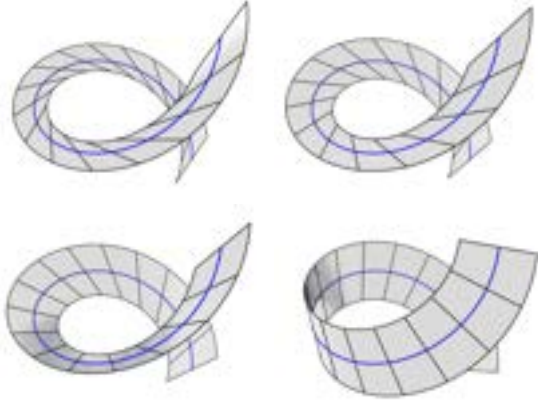


Fig. 12. Developable surfaces with constant tangent-ruling angle θ : From top left to bottom right, $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, and $\frac{\pi}{2}$. The helix is sampled with $N_{pt} = 200$.

5.2 Triply orthogonal structure

We've constructed a triply orthogonal structure using principles of differential geometry for developable strips [Gray 1999; Liu et al. 2006]. Starting with a helical developable strip around a helix with $\theta = \frac{\pi}{2}$ (RMF) and specific parameters ($a = 50$, $b = 25$, $w = 25$) as illustrated in the first image in Figure 1, we create a second helical strip orthogonal to the first one using (24) with $\varphi_0 = \frac{\pi}{2}$ as illustrated in the second image. The third strip is a plane orthogonal to both the first and second strips, remaining parallel to their respective rulings. The offset surface of a developable surface is another developable surface [Willmore 2012], and there are various methods to generate offset surfaces [Piegl and Tiller 1999]. In this approach, we generate offset points $\mathbf{op}_i$ and $\mathbf{om}_i$ using:

$$\mathbf{op}_i = \mathbf{p}_i + h\mathbf{N}_i, \quad \mathbf{om}_i = \mathbf{m}_i + h\mathbf{N}_i,$$

where h is the offset distance and $\mathbf{N}_i$ is the unit surface normal vector evaluated at $t = t_i$.

Using the fitting method from Section 3.4 for $\mathbf{op}_i$ and $\mathbf{om}_i$, we obtain B-spline curves $\mathbf{op}(t)$ and $\mathbf{om}(t)$. The offset developable strip is generated by linearly interpolating between $\mathbf{op}(t)$ and $\mathbf{om}(t)$, producing red (+ h) and yellow (− h) offset strips sandwiching the blue one, with an offset distance of $h = 25$ (see the third image). Similarly, by creating two offset surfaces on the second helical developable strips, we form a triply orthogonal structure comprised of these developable strips as shown in the fourth image. This process results in a triply orthogonal structure comprising these developable strips,

with lines of curvature tracing across the four offset surfaces as depicted in the last image.

5.3 Curve on a surface

The techniques presented in this section extend to curves on surfaces. Drawing inspiration from the Reichstag dome design, we shaped a hemisphere ($x^2 + y^2 + z^2 = a^2$) using a developable strip traced along a helix-like space curve within the hemisphere.

The parametric representation of this curve is expressed as

$$\mathbf{c}(t) = \left(\sqrt{a^2 - b^2 t^2} \cos(\lambda t), \sqrt{a^2 - b^2 t^2} \sin(\lambda t), bt \right), \quad (25)$$

where $0 \leq t \leq \frac{a}{b}$. To adapt to the widening intervals of the spiral curve toward the pole, we modulate the strip width, ensuring it increases along the parameter t .

The RMF representation of a curve on a sphere is given by $(\mathbf{t}, \mathbf{N} \times \mathbf{t}, \mathbf{N})$ [Wang et al. 2008], with $\mathbf{N}$ representing the unit surface normal of the sphere. Consequently, when using the RMF to generate a developable strip along a curve on a sphere, the normals of the sphere and the resulting developable strip seamlessly align as shown in Figure 13 (left). This alignment can also be achieved using the conventional method (16), which is equivalent to setting the rotation angle as $\varphi = \cos^{-1}(\mathbf{b} \cdot \mathbf{N})$. The parameters for the example depicted in Figure 13 (left) are $a = 1$, $b = 1$, and $\lambda = 108$.

When λ is reduced from 108 to 27, the strip widens, and the rulings start to self-intersect due to the emergence of the edge of regression as depicted in Figure 13 (center and right) blue strip. To address this issue, a parameter-dependent term $\Delta\varphi(t)$ is added to φ , ensuring its gradual increase along the parameter t (gray strip). This modification in φ is akin to the approach employed in the strip width case, effectively preventing self-intersection.



Fig. 13. Prevention of self-intersection is illustrated in the images. On the left, a developable surface (depicted in blue) is generated with $\lambda=108$ along a curve on a hemisphere (represented in yellow). In the center, when λ is reduced to 27, the blue strip, without modification, exhibits self-intersection, while the gray strip, which has been modified, successfully avoids self-intersection. The rightmost image provides a close-up view of the region marked by the black square in the center image. The directrix curve (25) is sampled with $N_{pt} = 500$.

6 Integrating the Darboux frame

In this section, we present an example for the technique presented in Section 3.2, where we compute the directrix curve and the Darboux frame of the developable strip simultaneously from given curvature and torsion data. Log-aesthetic curves, which have gained popularity in industrial design, exhibit a unique characteristic on a log-log scale: when plotting $\frac{\kappa'}{\kappa}$ against κ , they result in a straight line.

This property ensures that the curvature of these curves changes monotonically [Levien and Séquin 2009; Miura 2006]. The curvature of a log-aesthetic space curves, as described in [Yoshida et al. 2009], can be represented as:

$$\kappa(s) = \begin{cases} e^{-\Gamma s} & \alpha = 1 \\ (\Gamma\alpha s + 1)^{-\frac{1}{\alpha}} & \text{otherwise} \end{cases},$$

where $\Gamma = e^{-\bar{c}_1}$, and α denotes the slope and $\bar{c}_1$ represents the y-intercept of the straight line within the double logarithmic graph. The torsion for a monotonically increasing radius of torsion is given by:

$$\tau(s) = \begin{cases} e^{-(\Omega s + \log \nu)} & \beta = 0 \\ (\Omega\beta s + \nu\beta)^{-\frac{1}{\beta}} & \text{otherwise} \end{cases}.$$

Here, $\Omega = e^{-\bar{c}_2}$, and β and $\bar{c}_2$ are the slope and the y-intercept, respectively of the straight line in the double logarithmic graph, and ν is the radius of torsion at the reference point. The torsions for monotonically decreasing radius of torsion is given by:

$$\tau(s) = \begin{cases} e^{-(-\Omega s + \log \nu)} & \beta = 0 \\ (-\Omega\beta s + \nu\beta)^{-\frac{1}{\beta}} & \text{otherwise} \end{cases}.$$

Figure 14 illustrates the case where $\bar{c}_1 = 0.5$, $\bar{c}_2 = 0.5$, $\alpha = 2.0$, $\beta = 2.0$, $\nu = 1000$, and width of the strip is 4 ($w=2$). The rotation minimizing frame (24), described in Section 5, is used to compute the rotation angle φ .

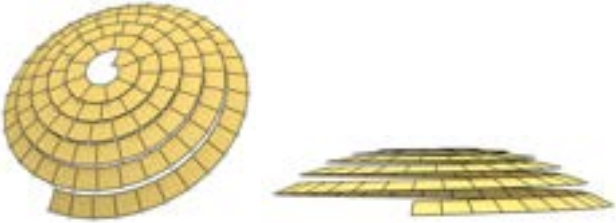


Fig. 14. Developable surfaces associated with log aesthetic curves. Left: perspective view with control points represented as circles, Right: front view. The number of sample points used in the integration interval is $N_{pt} = 600$.

7 Applications

We offer illustrative examples that focus on controlling the width of developable strips, designing windmill blades, exploring curved folding, and creating a model that integrates various techniques developed in this study.

7.1 Strip width control

In this section, we demonstrate how adjusting the width of developable strips can significantly enhance their usefulness in real-world applications using two models.

7.1.1 Deltoid. The deltoid (or tricuspid) is a curve traced by a point on the circumference of a circle rolling inside a circle three times as large [Lawrence 1972]. The parametric equation is given by

$$\mathbf{c}(t) = (a(2 \cos(t) + \cos(2t)), a(2 \sin(t) - \sin(2t)), 0),$$

where $0 \leq t \leq 2\pi$ and a is the radius of the small rolling circle. When viewed from the top, we construct an evolute of the deltoid, which interestingly forms another deltoid as illustrated in Figure 15 (left). This construction is expressed in terms of developable surfaces along the deltoid curve. We assume that $\varphi(t)$ is a constant, denoted as $\varphi = \varphi_{del}$. Consequently, $\tau_g = 0$, leading to $\mathbf{d} \cdot \mathbf{t} = 0$. The plane of the deltoid is spanned by $(\mathbf{t}, \mathbf{n})$, and the projection of the ruling direction $\mathbf{d}$ of the developable strip into this plane is expressed as

$$\mathbf{d}_{proj} = (\mathbf{d} \cdot \mathbf{t})\mathbf{t} + (\mathbf{d} \cdot \mathbf{n})\mathbf{n} = \cos(\varphi_{del})\mathbf{n}.$$

Thus, the length of the ruling $w(t)$ can be obtained as

$$w(t) \cos(\varphi_{del}) = \frac{1}{\kappa(t)}$$

where $\kappa(t) = \frac{-1}{8a \sin(1.5t)}$. Figure 15 illustrates the top, front, and perspective views of the three developable strips constructed around the deltoid.



Fig. 15. Deltoid evolute: From left to right, Top view, right view, and perspective view. One-third of the deltoid curve is sampled with $N_{pt} = 300$.

7.1.2 Loxodrome. A loxodrome is a curve on a sphere that intersects all meridians at the same angle α [do Carmo 1976]. Denoting the sphere as

$$x(u, v) = r \sin u \cos v, \quad y(u, v) = r \sin u \sin v, \quad z(u, v) = r \cos u,$$

where $0 \leq u \leq \pi$ and $0 \leq v \leq 2\pi$, the equation of the loxodromes is given by

$$v = a \ln \tan \left(\frac{u}{2} \right) + c, \quad (26)$$

with $a = \tan \alpha$ and c as the integration constant [do Carmo 1976]. The parameter c shifts the loxodrome curve in the v direction. Solving (26) for u , we get

$$u = 2 \arctan \left(e^{\frac{v-c}{a}} \right). \quad (27)$$

We can generate n loxodrome curves by shifting c by intervals of $\frac{2\pi}{n}$. Using the RMF technique, we can create the rulings of the developable surface along the loxodrome. The width of the strip along parameter v is given by

$$w(v) = \cos \left(\frac{\pi}{2} - \alpha \right) \frac{r}{2} (u_b - u_a), \quad (28)$$

where $u_a = 2 \arctan \left(e^{\frac{v-c_a}{a}} \right)$ with $c_a = \frac{2\pi}{n} - \frac{\pi}{n}$ and $u_b = 2 \arctan \left(e^{\frac{v-c_b}{a}} \right)$ with $c_b = \frac{2\pi}{n} + \frac{\pi}{n}$.

The example illustrated in Figure 16 consists of four equal developable strips ($n = 4$) to approximate the sphere's shape. For this model, the maximum and average deviations from the sphere's diameter are computed to be 0.629% and 0.327%, respectively. The top

row, from left to right, displays the top, front, and perspective views. In contrast, the bottom row, also from left to right, showcases the top, front and perspective views of the manually constructed model from these strips. The four unfolded strips are shown in the center. It's worth noting that the papercraft model has an opposite rotation, as the rear sides of the paper strips are utilized.



Fig. 16. Loxodrome: Top left to bottom - top view, front view, perspective view, the unfolded shape, the three respective views of the papercraft model. Each loxodrome is sampled with $N_{pt} = 1000$.

7.2 Windmill blades

7.2.1 Toroidal blades. Taking inspiration from MIT's toroidal propeller concept [Sebastian and Strem 2019], we have created a model resembling it with a developable surface. This model is based on an ellipse with a semi-major axis of 50 mm and a semi-minor axis of 45 mm, $w = 20$ mm, and $\varphi(t) = 0.8 \cos(t) + \frac{\pi}{2}$, thus, $\frac{d\varphi}{ds} = \frac{1}{\Lambda(t)} \frac{d\varphi}{dt} = -\frac{0.8 \sin(t)}{\Lambda(t)}$. Figure 17 shows, top left to bottom right, perspective view, top view, front view, and the papercraft model of the toroidal developable surface. The unfolded shape is presented in the bottom right image in Figure 5. Inserting a 2mm aluminum pipe into the toroidal blade, as depicted in Figure 17 bottom right, makes the blade rotate orthogonally to the pipe axis when the wind blows parallel to that axis. The corresponding video is available in the supplemental material.

7.2.2 Helical blades. In contrast to horizontal-axis wind turbines, vertical-axis wind turbines are inherently designed to be insensitive to the wind direction, and thus do not require a yaw control system [Han et al. 2018]. Furthermore, many commercial vertical-axis wind turbines incorporate a design in which the blades are helically twisted around the turbine's rotational axis [Han et al. 2018].

Based on this information, we have created a papercraft windmill model with $a = 50$, $b = 35$, $0 \leq t \leq \frac{2\pi}{3}$ consisting of three helical

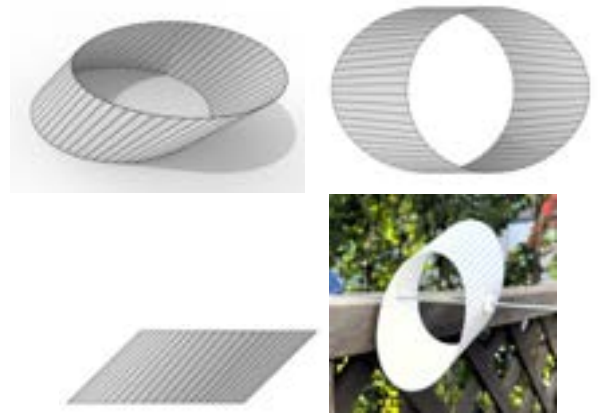


Fig. 17. Toroidal blades: From top left to bottom right, perspective view, top view, front view, and the papercraft model. The ellipse is sampled with $N_{pt} = 300$.

developable blades using our technique. The rotation angle $\varphi(t)$ is obtained such that the tangent of the helix and the rulings are at right angles, i.e. $\theta = \frac{\pi}{2}$ (RMF). Furthermore, we set $\varphi_0 = \frac{\pi}{2}$ so that the blades obtain lift rather than drag from the air. In Figure 18, the top left and top right images show the perspective and top views, respectively, illustrating the arrangement of three blades spaced at intervals of 120° . The unfolded representation of the helical blade is depicted in the bottom left, while the assembled papercraft model can be seen in the bottom right. The video in the supplemental material shows its ability to operate as a horizontal-axis wind turbine as well.

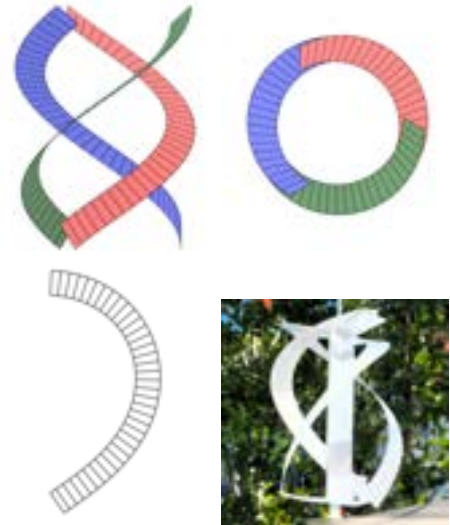


Fig. 18. Helical windmill: From top left to bottom right, perspective view, top view, unfolded shape, and the papercraft model. The helix is sampled with $N_{pt} = 100$.

7.3 Curved folding

We've created a curved folding model consisting of two interconnected developable strips along a helical crease. Our process begins by determining the rotation angle $\varphi(t)$, through the solution of an initial value problem using the differential equation (24) with a negative sign which coincides with Eqn. (4) of [Tachi 2011]. In this case, utilizing the differential equation with a positive sign fails to satisfy the condition. In this specific example, we maintain a fixed angle of $\theta = \frac{\pi}{4}$ between the tangent and the ruling direction. The top left graph in Figure 19 provides the solution $\varphi(t)$ to the initial value problem, graphed against the curve parameter t .

Utilizing the solution φ_i , ($i = 0, \dots, n$) as a basis, we can generate four distinct developable surfaces through the evaluation of $\mathbf{d}$ at various angles: $\mathbf{d}(\varphi_i)$, $\mathbf{d}(\pi - \varphi_i)$, $\mathbf{d}(-\varphi_i)$, and $\mathbf{d}(\pi + \varphi_i)$. This process results in the generation of developable surfaces denoted as D_{top}^+ , D_{top}^- , D_{bot}^+ , and D_{bot}^- , as illustrated in the top right of Figure 19. The upper pair, D_{top}^+ and D_{top}^- , as well as the lower pair, D_{bot}^+ and D_{bot}^- form curved foldings. The middle left image depicts the perspective view of the resulting lower pair of curved folding along the helix. The middle right image is its top view.

For developable strips with a curved fold, there always exist two developable strips sharing the crease curve. Thus, implementing the method introduced in Section 3.5 is more involved as we need to flatten each side of the developable surface separately and then combine them. Knowing that the plus and minus sides of the developable strip share the same geodesic curvature $\kappa_g = \kappa \cos(\varphi) = \kappa \cos(-\varphi)$ along the crease curve, we first unfold it by solving the ordinary differential equation $\mathbf{t}' = \kappa \mathbf{n}$ using $\kappa = \kappa_g$ and $\mathbf{n} = (0, 0, 1) \times \mathbf{t}$. As the angles between the tangent of the crease curve and the ruling direction of both sides remain unchanged along the curve during the unfolding process, unfolding the rulings to both sides isometrically becomes a straightforward task. The bottom left of Figure 19 illustrates the unfolded result of the curved folded model, while the bottom right depicts the folded papercraft model.

7.4 Sea shell strips

A sea shell surface is generated as a circular tube wrapped around a helix, with the radius of the circle varying proportionally to the helix's parameter as follows:

$$\mathbf{r}(t, \theta) = \text{helix}(t) + rt(-\cos(\theta)\mathbf{n}(t) + \sin(\theta)\mathbf{b}(t)), \quad (29)$$

where r is a constant. We construct two developable strips along $\theta = 0$ and π of a sea shell surface with $a = 20$, $b = 30$ for the helix, and $r = 15$. The half-width w is set to be proportional to the parameter t as $\frac{rt}{8}$. The rotation angle is defined as $\varphi(t) = p \sin(t) + q$. Initially, we set $p = 0.5$ and $q = \frac{\pi}{6}$; however, this configuration causes self-intersection due to the emergence of the edge of regression, as illustrated in the top left of Figure 20. To resolve this, we increased q from $\frac{\pi}{6}$ to $\frac{\pi}{4}$, as shown in the top center of Figure 20, where both developable strips are displayed together to highlight the angle difference at the top of the strips. The top right image shows the result with $q = \frac{\pi}{4}$, without self-intersection.

For the developable strip with $\theta = \pi$, we set p to -0.5 and q to $-\frac{\pi}{4}$. We further refined the edge curves of both developable strips by incorporating sinusoidal modifications. The added sinusoidal curve

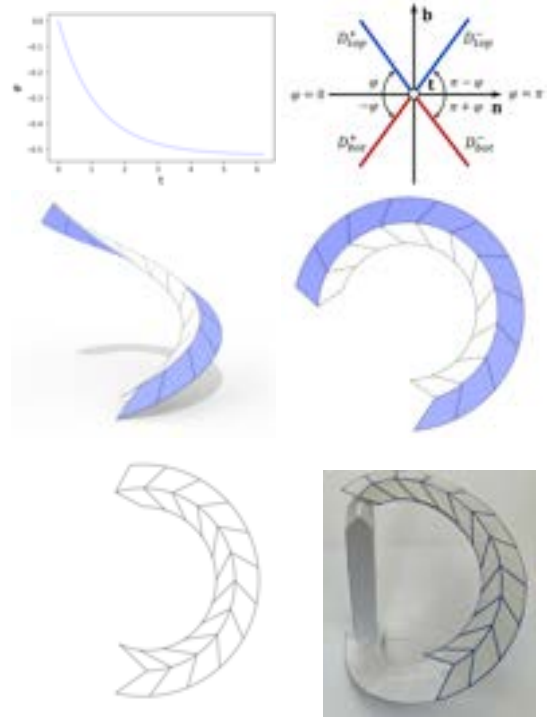


Fig. 19. Curved folding: From top left to bottom right, graph of $\varphi(t)$, definition of four developable strips D_{top}^+ , D_{top}^- , D_{bot}^+ , D_{bot}^- , perspective and top views of the bottom pair of curved folded model, flattened model, and the papercraft model. The helix is sampled with $N_{pt} = 100$.

is defined as $w = \frac{rt}{8}(1 + 0.2 \sin(30t))$, and the minus edge curve is defined as $w = \frac{rt}{8}(1 + 0.2 \cos(30t))$. The top, front, and perspective views of the $\theta = 0$ and π strips are illustrated in the bottom row of Figure 20.

7.5 Comparison with previous methods

In this section, we compare our method with those proposed by [Bo and Wang 2007], and [Raffaelli et al. 2018] for constructing developable strips.

[Bo and Wang 2007] introduced a method for modeling developable surfaces to simulate paper bending in interactive and animation applications. Their approach is based on the concept that, for a space curve with non-vanishing curvature, the envelope of its rectifying planes forms a rectifying developable surface, with the curve itself being a geodesic on this surface. Figure 21, in the top row, shows a front and right view of a rectifying developable strip generated along a cubic Bézier curve defined by the control points $(0, 0, 0)$, $(1, 2, 1)$, $(2, 3, 4)$ and $(4, 0, 6)$. While the method presented by [Bo and Wang 2007] is limited to rectifying developables where $\varphi = \frac{\pi}{2}$, our method is more versatile, treating the rectifying developable as just one possibility. As shown in Figure 21, our approach also accommodates $\varphi = \frac{\pi}{4}$ (middle row) and the RMF option (bottom left: front view and bottom center: right view). The right view of the RMF option clearly demonstrates that the generators are orthogonal



Fig. 20. Sea shell surface: From top left to bottom right, the images show a developable strip with $q = \frac{\pi}{6}$ exhibiting self-intersection, developable strips with $q = \frac{\pi}{6}$ and $q = \frac{\pi}{4}$ depicted together to highlight the angle difference at the top of the strips, a developable strip with $q = \frac{\pi}{4}$ without self-intersection, and the top, front, and perspective views of the $\theta = 0$ and π strips. The directrix curve (29) is sampled with $N_{pt} = 400$.

to the directrix curve. Finally, the bottom right image shows the three developable strips sharing the same cubic Bézier curve. These results clearly highlight the broader applicability and flexibility of our approach.

[Raffaelli et al. 2018] present a method to approximate a surface using multiple developable strips (ribbons), which they call ribbonization, by employing the conventional method described in Section 3.1.1. They illustrate this method with an example using a torus with a major radius R and minor radius r , where the aspect ratio $\frac{r}{R}$ is 0.5. The example features two ribbons along helical curves on the torus, parameterized as follows:

$$\mathbf{c}(t) = ((R + r \cos(\omega t)) \cos t, (R + r \cos(\omega t)) \sin t, r \sin(\omega t)) ,$$

and

$$\mathbf{c}(t) = \begin{pmatrix} (R + r \cos(\omega t)) \cos \left(t + \frac{\pi}{\omega}\right) \\ (R + r \cos(\omega t)) \sin \left(t + \frac{\pi}{\omega}\right) \\ r \sin(\omega t) \end{pmatrix}^T$$

where $\omega = 3$ is used. Raffaelli et al.'s method is limited to generating developable strips using the conventional approach, where the rotation angle $\varphi(t)$ is set to $\varphi = \cos^{-1}(\mathbf{b} \cdot \mathbf{N})$, as shown by the blue curve in the graph at the top of Figure 22. In contrast, our method is more versatile, setting $\varphi(t) = p \sin(\omega t) + q$, where the amplitude p remains the same as in the conventional method, but q is increased by 20% as shown by the red curve and decreased by 20% as shown

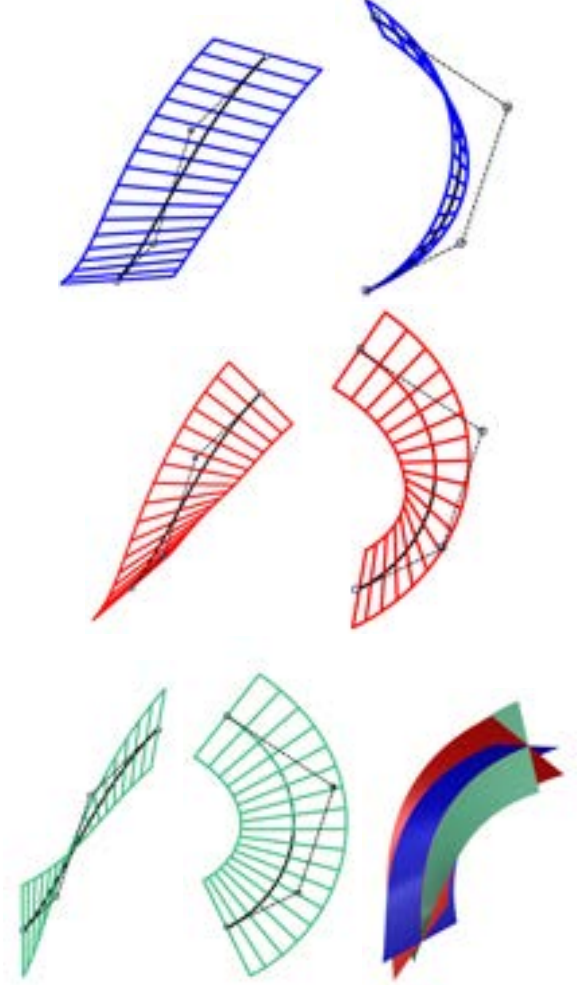


Fig. 21. Generation of three developable strips sharing the same cubic Bézier curve. From top left to bottom right: $\varphi = \frac{\pi}{2}$ (front view and right view), $\varphi = \frac{\pi}{4}$ (front view and right view), RMF (front view and right view), and three developable strips sharing the same Bézier curve. Each developable strip is sampled with $N_{pt} = 100$.

by the green curve in the graph. Figure 22 depicts the corresponding ribbonizations.

Figure 23 depicts a ribbonization of a torus with an aspect ratio of 0.205, created using two triply orthogonal structures with $\omega = 7$. For general aspect ratios and values for ω , the in-plane orientations of the square cross-sections of the triply orthogonal structure at both ends of the directrix curve do not coincide due to the rotation angle $\varphi(t)$, which is determined by the rotation minimizing frame. The angles at either side of the joint need to be equal up to an integer multiple of 2π to ensure alignment. To satisfy this condition, we adjust the aspect ratio of the torus. For $\omega = 7$, the initial aspect ratio of 0.2, which fails to align as shown in Figure 24 (left), was refined to 0.205 for proper alignment, as illustrated in Figure 24 (right), using the bisection method. These examples clearly demonstrate

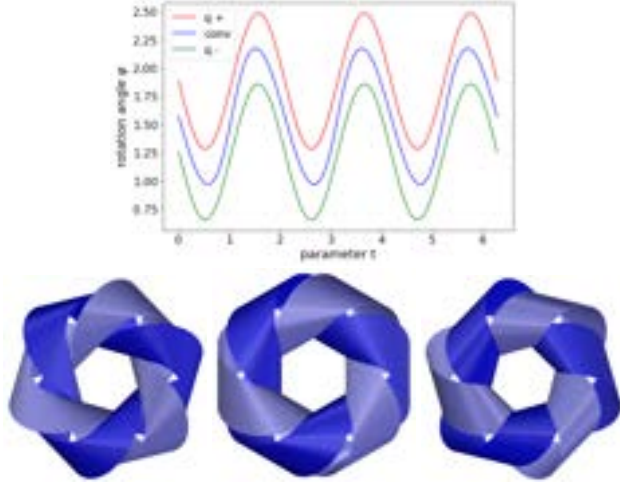


Fig. 22. Comparison of our method with that of [Raffaelli et al. 2018]. Top: Graphs of $\varphi(t)$ for three cases: q increased by 20% ($q+$) in red, the conventional method (conv) in blue, and q decreased by 20% ($q-$) in green. Bottom: the ribbonizations corresponding to each case (left: $q-$, middle: conv, right: $q+$), with each ribbon sampled at $N_{pt} = 1000$.



Fig. 23. Comparison of our method with that of [Raffaelli et al. 2018]: Ribbonization of a torus with an aspect ratio of 0.205, constructed using two triply orthogonal structures with $\omega = 7$. Each developable strip in the triply orthogonal structure is sampled at $N_{pt} = 1000$.

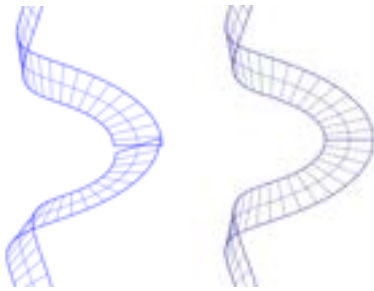


Fig. 24. Alignment of both ends of the developable strip constructed using RMF: The aspect ratio of the torus is adjusted to meet the alignment condition. For $\omega = 7$, the initial aspect ratio of 0.2, which does not align (left), was refined to 0.205 for proper alignment (right) using the bisection method.

the importance of the newly introduced method for controlling the rotation angle $\varphi(t)$ compared to existing methods.

8 Conclusion and future work

In this study, we introduced a novel method for constructing developable surfaces along space curves including curves on surfaces. The method is based on various ways of controlling the rotation angle φ between the Frenet frame of the space curve and its Darboux frame as a curve on the resulting developable surface. We showed a wide range of applications, spanning fields such as architecture, windmill blade design for papercraft models, curved folding, triply orthogonal structures, and the creation of surfaces featuring a log-aesthetic curve.

Although this study has made significant strides, several challenges still lie ahead. One of the key challenges is exploring more analytical functions and polynomial curves for rotation angles. Furthermore, our future endeavors involve representing the rotation angle φ and the directrix curve c using an artificial neural network, and then optimizing this network with respect to an objective function. We aim to use these findings to approximate free-form surfaces with a number of developable strips. We expect that the increased flexibility of our new approach will enable the approximation of more complicated surfaces that cannot be handled by the existing approaches for approximation with developable strips.

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